CONCRETE EVIDENCE

Causally-Grounded Diagnosis for Construction Delay Prediction

Stage 1 Project Proposal

Problem

Construction disputes are decided by forensic schedule analysts who manually reconstruct causal chains — for example, a differing site condition leading to a design freeze, then a weather delay, then a critical-path impact — by working through thousands of pages of RFIs, change orders, and correspondence. Every existing "AI delay prediction" tool does the opposite: it trains a flat tabular model on resource scores and cost ratios and outputs a risk percentage with no causal reasoning behind it. No existing system connects the two. The open question we chose to pursue: when a delay-prediction model gets it wrong, can real legal precedent explain why — and eventually help it get it wrong less often?

Proposed Solution

CONCRETE EVIDENCE is a two-stage pipeline that cross-checks a predictive delay-risk model's failures against real construction-dispute case law. It is deliberately scoped as an ideation and validation phase: the goal for this stage is to prove that a diagnostic, precedent-grounded layer works before building the closed-loop system that uses it to actually improve predictions.

Phase 1 — What We Have Built and Validated

The predictor. An XGBoost classifier trained on a 1,301-row construction task dataset, engineered into 14 features (resource constraint, site constraint, cost-per-labor, dependency count, and others), with full SHAP explainability attached to every prediction.

The forensic layer. 22 real U.S. construction delay disputes (ASBCA appeals and federal court opinions) pulled from CourtListener, scored by an LLM for causal richness, then parsed into structured causal DAGs. Nodes represent events (differing site conditions, weather delay, liquidated-damages trigger, and similar); edges represent typed relationships (caused, contributed to), each tagged with an evidentiary confidence tier so that a judge's finding of fact is weighted differently from an unproven allegation. 16 of the 22 cases yielded chains rich enough to extract.

The alignment layer. A single taxonomy crosswalks the predictive model's feature names to the forensic layer's event types, so the two independently built systems can communicate.

The cross-check. For every misclassified test-set task, the pipeline takes the top SHAP-driving features, maps them through the taxonomy to forensic event types, queries the case database, ranks matches, and walks the matched case's causal graph outward from the anchor node. The output is a plain-English diagnosis: not just that the model was wrong, but which real legal precedent shows the causal cascade it missed.

Validation result: on a 260-task test set, the model produced 28 misclassifications (10.77% error rate), and 27 of those 28 (96.4%) were matched to a real forensic precedent explaining the gap. This validates the core thesis — flat-feature models miss multi-hop causal cascades, and real case law can pinpoint exactly where.

Phase 2 — What We Are Building Next

The forensic layer is currently read-only: it explains the model's blind spots after the fact but does not yet fix them. That is the actual research question this project is scoped to answer next, in priority order:

- Close the loop, not just diagnose it. Engineer causal-chain features directly from the DAGs (e.g., does this task's active event types appear upstream of a critical-path impact in precedent, or what is the matched-precedent liquidated-damages rate) and feed them back into the predictor — turning a diagnosed blind spot into a measurably better model.
- Make the diagnosis self-triggering, so the system automatically flags which feature is structurally blind and proposes the feature that closes the gap, rather than a human reading a static report.
- Scale the precedent base beyond 22 cases and track a coverage-gap metric as it grows, and validate the LLM-assigned causal-confidence tiers against manual review — rigor that matters if this ever touches real claims work.
- Run it on live, unseen projects through a real intake path that takes a folder of schedules, RFIs, or change orders for an active project and outputs a prediction with an evidence-backed causal explanation, rather than only a static test split.
- Visualize the causal chain, since the DAG walk is the most compelling artifact the system produces and it is currently buried in a markdown report.

Chosen Track

Track 4: Open Innovation